Progression of Our Automations using Python

```
Python Automations
|
├── 1. Introduction to Automation
|   ├── What is Automation?
|   ├── Benefits & Use Cases (files, web, emails, reports)
|   └── Required Libraries & Setup (pip, venv, .env for secrets)
|
├── 2. System & File Automations
|   ├── Automating File Ops (os, shutil, pathlib)
|   ├── Bulk rename/move/copy/delete
|   ├── Directory sync & backup scripts (hash compare, timestamps)
|   └── Error handling & retries (try/except, backoff)
|
├── 3. Scheduling & Orchestration
|   ├── Periodic Tasks (schedule)
|   ├── OS Schedulers
|   ├── Long-running daemons & services (systemd, NSSM)
|   └── Dependency chains
|
├── 4. Text & Document Automations
|   ├── Bulk search/replace
|   ├── CSV/Excel processing
|   └── Automated report generation
|
├── 5. Email Automations
|   ├── Send emails (smtplib, yagmail) with HTML & attachments
|   └── Notification systems (success/failure alerts, daily summaries)
|
├── 6. OS & Process Automations
|   ├── Run system commands (subprocess.run, capture output)
|   ├── System monitoring (CPU, memory, disk, net) with psutil
|   ├── Process info: list processes, name, PID, CPU%, RSS/Memory (psutil)
|   ├── Start/stop/restart processes safely
|   └── Log file monitoring & tailing
|
├── 7. Concurrency for Automation
|   ├── Multithreading (I/O-bound tasks: downloads, API calls)
|   ├── Multiprocessing (CPU-bound: parsing, compression, transforms)
|   ├── Process pools & queues (concurrent.futures, multiprocessing.Pool)
|
├── 8. Logging, Auditing & Maintenance
|   ├── logging basics (levels, formatters, handlers)
```

Progression of our Data Science Journey

```
Artificial Intelligence (AI)

├── 1. Data Science

│   ├── Data Collection

│   │   ├── Databases (SQL, NoSQL)

│   │   ├── APIs, Web Scraping

│   │   ├── Streaming Data (Kafka, IoT devices)

│   │   └── Open Datasets (Kaggle, UCI Repository)

│   │

│   ├── Data Cleaning (ETL)

│   │   ├── Handling Missing Values

│   │   ├── Outlier Detection & Treatment

│   │   ├── Data Transformation (scaling, encoding)

│   │   └── Feature Engineering

│   │

│   ├── Exploratory Data Analysis (EDA)

│   │   ├── Descriptive Statistics

│   │   ├── Correlation Analysis

│   │   ├── Hypothesis Testing

│   │   └── Data Summaries & Patterns

│   │

│   ├── Data Visualization

│   │   ├── Charts (Bar, Pie, Line, Scatter)

│   │   ├── Heatmaps & Histograms
```

```
|   |   ├── Dashboards (Tableau, Power BI)
|   |   └── Python Libraries (Matplotlib, Seaborn, Plotly)
|   |
|   └── Applied AI + Statistics
|       ├── Probability & Distributions
|       └── Decision Making with Data
|
├── 2. Machine Learning (ML)
|   ├── Supervised Learning
|   |   ├── Regression
|   |   |   ├── Linear Regression
|   |   |   ├── Logistic Regression
|   |   |   └── Regularization
|   |   |
|   |   ├── Classification
|   |   |   ├── K-Nearest Neighbors (KNN)
|   |   |   ├── Support Vector Machines (SVM)
|   |   |   ├── Decision Trees
|   |   |   ├── Random Forests
|   |   |
|   |   └── Ensemble Methods
|   |       ├── Bagging
|   |       ├── Boosting
|   |
```

```
|   ├── Unsupervised Learning
|   |   ├── Clustering
|   |   |   ├── K-Means
|   |   |   └── DBSCAN
|   |   └── Dimensionality Reduction
|   |       ├── PCA (Principal Component Analysis)
|
└── 3. Deep Learning (DL)
    ├── Neural Networks (NN)
    |   ├── Basics of Neurons & Perceptrons
    |   ├── ANN (Artificial Neural Networks)
    |   ├── CNN (Convolutional Neural Networks)
    |   |   ├── Image Classification
    |   |   ├── Object Detection
    |   |   └── Transfer Learning
    |   ├── RNN (Recurrent Neural Networks)
    |   |   ├── LSTM (Long Short-Term Memory)
    |   |   ├── GRU (Gated Recurrent Unit)
    |   |   └── Applications: Text, Speech, Time Series
    |   └── Autoencoders
    |       ├── Dimensionality Reduction
    |       ├── Anomaly Detection
    |       └── Generative Autoencoders (VAE)
    |
```

```
├── Transformers (Attention-based Architecture)
│   ├── Key Concepts
│   │   ├── Self-Attention
│   │   ├── Multi-Head Attention
│   │   ├── Positional Encoding
│   │   └── Encoder/Decoder Blocks
│   │
│   ├── Model Types
│   │   ├── Encoder-only (BERT → Text Understanding)
│   │   ├── Decoder-only (GPT → Text Generation)
│   │   └── Encoder–Decoder (T5, BART → Summarization, Translation)
│
├── LLMs (Large Language Models)
│   ├── Pretraining
│   │   ├── Next Word Prediction
│   │   └── Masked Language Modeling
│   ├── Fine-tuning
│   │   ├── Domain Adaptation
│   │   └── Instruction Tuning
│   ├── RLHF (Reinforcement Learning with Human Feedback)
│   ├── Multimodal LLMs
│   │   ├── Text + Image (GPT-4V, Gemini)
│   │   ├── Text + Audio (Whisper + LLMs)
```

```
|   |   └── Text + Video (emerging research)
|   └── Applications
|       ├── Chatbots (ChatGPT, Claude, Gemini)
|       ├── Code Generation (Copilot, Code Llama)
|       ├── Content Generation (blogs, emails, reports)
|       ├── Summarization / Translation
|       └── Decision Support Systems
```

Progression of Python Programming Journey

```
Python Programming
│
├── 1. Introduction & Basics
│   ├── History and Features of Python
│   ├── Installation (Python + IDE setup)
│   ├── First Program ("Jay Ganesh")
│   └── Python Syntax, Indentation, Comments
│
├── 2. Core Data Types
│   ├── Numbers (int, float, complex)
│   ├── Strings (creation, slicing, formatting)
│   ├── Booleans
│   └── Type Casting & Input/Output
│
├── 3. Sequence Data Types
│   ├── List
│   │   ├── Indexing, Slicing
│   │   ├── Methods (append, insert, pop, sort, etc.)
│   ├── Tuple
│   ├── Set (union, intersection, difference)
│   └── Dictionary (keys, values, CRUD operations)
│
├── 4. Operators & Expressions
│   ├── Arithmetic Operators
│   ├── Relational Operators
│   ├── Logical Operators
│   ├── Membership & Identity Operators
│   └── Operator Precedence
│
├── 5. Control Flow
│   ├── Conditional Statements (if, if-else, if-elif-else)
│   ├── Loops
│   │   ├── for loop
│   │   ├── while loop
│   │   └── Nested loops
│   ├── break, continue, pass
│   └── Iterators & Generators (basic intro)
│
├── 6. Functions & Modular Programming
│   ├── Defining Functions
│   ├── Arguments (positional, keyword, default, variable length)
│   ├── Return values
│   ├── Lambda Functions
│   ├── Recursion
│   ├── Modules (import, user-defined modules)
│   └── Command-line Arguments
│
├── 7. Strings (Advanced)
```

```
│   ├── String Methods
│   ├── String Formatting
│   └── Regular Expressions (re module)
│
├── 8. File Handling
│   ├── Opening/Closing Files
│   ├── Read/Write/Append
│   ├── File Modes
│   └── Exception Handling with Files
│
├── 9. Object-Oriented Programming (OOP)
│   ├── Classes & Objects
│   ├── __init__ Constructor
│   ├── Methods (instance, static, class methods)
│   ├── Inheritance (single, multiple, multilevel)
│   ├── Polymorphism & Overriding
│   ├── Encapsulation & Abstraction
│
├── 10. Exception Handling
│   ├── try, except
│   ├── else, finally
│   ├── Custom Exceptions
│   └── Raising Exceptions
│
├── 11. Functional Programming Tools
│   ├── map(), filter(), reduce()
│   ├── Decorators (basic & advanced)
│   └── Generators & yield
│
├── 12. Libraries & Packages (Essentials)
│   ├── math, random, datetime
│   ├── OS and sys
│   ├── collections
│   └── pip & virtual environments
│
├── 13. Python with Data
│   ├── NumPy (arrays, operations)
│   ├── Pandas (Series, DataFrames, CSV handling)
│   ├── Matplotlib (basic plotting)
│   └── Introduction to Seaborn
│
└── 14. Advanced Concepts (Optional / Bridge to ML)
    ├── Exception handling in real projects
    ├── Introduction to Automation (os, shutil, subprocess)
    ├── Multithreading & Thread Synchronization
    ├── Multiprocessing & Process Pooling
    ├── Multicore Programming Concepts
    ├── Advanced Decorators (function wrappers)
    ├── Intro to ML Libraries (scikit-learn, TensorFlow)
```

Python Career Streams
&
Industry Opportunities

Data Science

- **Core stack:** numpy, pandas, matplotlib/plotly, seaborn, scikit-learn, statsmodels, Jupyter, mlflow (tracking), great_expectations (data quality).
- **Concepts:** EDA, feature engineering, train/validation/test splits, cross-validation, metrics (RMSE/MAE, accuracy/F1/AUC), leakage avoidance, bias/variance, model explainability (SHAP).
- **Roles:** Data Analyst, Data Scientist.
- **Resume keywords:** EDA, feature engineering, cross-validation, SHAP, MLflow, reproducible pipeline.

Web Backends & APIs

- **Core stack:** FastAPI/Django (ORM: SQLAlchemy/Django ORM), PostgreSQL/MySQL, Redis, JWT/OAuth2, Celery/RQ, Docker.
- **Concepts:** REST, async I/O, pagination, auth, caching, background jobs, testing (pytest), CI/CD.
- **Roles:** Python/Backend Developer, API Engineer.
- **Resume keywords:** FastAPI/Django, ORM, JWT, Redis caching, Celery, Docker, CI/CD.

Data Engineering & Pipelines

- **Core stack:** Airflow/Prefect (orchestrate DAGs), PySpark/Dask (big data), Kafka, dbt, Parquet/Delta, S3/GCS, DuckDB.
- **Concepts:** Batch vs streaming, partitioning, schema evolution, lineage, SLAs, data tests, CDC.
- **Roles:** Data Engineer, ETL Developer.
- **Resume keywords:** Airflow DAGs, PySpark, Kafka, Parquet, dbt models, data quality.

DevOps/SRE & Cloud Automation

- **Core stack:** AWS Boto3/Azure/GCP SDKs, Terraform/Pulumi (Python), Kubernetes Python client, GitHub Actions, Prometheus/Grafana.
- **Concepts:** IaC, blue-green/canary, observability (logs/metrics/traces), drift detection, cost controls.
- **Roles:** DevOps Engineer, SRE, Cloud Engineer.
- **Resume keywords:** IaC (Terraform/Pulumi), Kubernetes client, CI/CD, Boto3, observability.

Cybersecurity & DFIR

- **Core stack:** scapy, python-nmap, cryptography, yara-python, volatility3, paramiko.
- **Concepts:** Network scanning, log parsing, IOC rules, secure key management, forensics triage.
- **Roles:** Security Analyst, DFIR Engineer (add certs: Sec+, eJPT, etc.).
- **Resume keywords:** IOC detection, YARA, Volatility, secure crypto, SSH automation.

IoT/Embedded & Edge

- **Core stack:** MicroPython/CircuitPython (ESP32), Raspberry Pi (gpiozero), MQTT (paho-mqtt), OpenCV, TF-Lite.
- **Concepts:** Sensor I/O, pub/sub messaging, edge inference, reliability & watchdogs, OTA updates (basic).
- **Roles:** IoT/Embedded Engineer, Robotics (entry).
- **Resume keywords:** MicroPython, MQTT, Raspberry Pi, edge inference, OpenCV.

Artificial Intelligence (AI)

- **Definition:** Making machines that can think, reason, and act like humans.
- **Relation:** The **umbrella field** under which everything else falls.

Data Science

- **Definition:** Extracting knowledge and insights from data using AI + statistics.
- **Relation:** Uses AI/ML tools + statistics for solving real-world problems.

Sub-steps:

- **Data Collection** → Gathering raw data (sensors, websites, databases).
- **Data Cleaning (ETL)** → Removing errors, filling missing values, transforming formats.
- **EDA (Exploratory Data Analysis)** → Finding trends and patterns in data.
- **Data Visualization** → Graphs, dashboards to make data easy to understand.
- **Applied AI + Statistics** → Using ML/AI methods for predictions/decisions.

Machine Learning (ML)

- **Definition:** A method in AI where machines **learn patterns from data** without being explicitly programmed.
- **Relation:** A **subset of AI**, forms the core of modern AI systems.

Types:

- **Supervised Learning** → Learning from labeled data.
  - **Regression** → Predict numbers (e.g., house price).
  - **Classification** → Predict categories (spam/not spam).
  - **Ensemble Methods** → Combine many models for better accuracy (Random Forests, XGBoost).
- **Unsupervised Learning** → No labels, find hidden patterns.
  - **Clustering** → Group similar items (customer segmentation).
  - **Dimensionality Reduction** → Compress data but keep info (PCA, t-SNE).
- **Reinforcement Learning (RL)** → Learn by trial and error, with rewards/punishments.
  - **Q-Learning, DQN, Policy Gradient** → Used in games, robotics.

- **Feature Engineering / Evaluation** → Selecting good inputs & testing models.

Deep Learning (DL)

- **Definition:** A subfield of ML that uses **neural networks with many layers**.
- **Relation:** Advanced ML → powerful for images, speech, text.

Neural Networks:

- **ANN (Artificial Neural Networks)** → Basic model (input → hidden → output).
- **CNN (Convolutional Neural Networks)** → Best for images, detect features like edges/faces.
- **RNN (Recurrent Neural Networks)** → Best for sequential data like text/speech.
  - **LSTM (Long Short-Term Memory)** → RNN with better memory.
  - **GRU (Gated Recurrent Unit)** → Lighter alternative to LSTM.
- **Autoencoders** → Compress data, detect anomalies.

Other Architectures:

- **GANs (Generative Adversarial Networks)** → Two models compete (Generator creates, Discriminator judges). Used in fake image generation (DeepFakes).
- **Transformers** → Use **self-attention**, process whole sequences in parallel.
  - **Encoder-only (BERT)** → Understanding text.
  - **Decoder-only (GPT)** → Generating text.
  - **Encoder–Decoder (T5, BART)** → Translation, summarization.

LLMs (Large Language Models)

- **Definition:** Very large Transformer models trained on massive text data.
- **Relation: Built using Transformers**, scaled up with billions of parameters.

Steps:

- **Pretraining** → Learn general language patterns (predict missing/next word).
- **Fine-tuning** → Specialize for tasks (medical chatbot, coding assistant).
- **RLHF (Reinforcement Learning with Human Feedback)** → Align with human expectations (safe, polite).

- **Multimodal LLMs** → Handle text + image + audio (e.g., GPT-4, Gemini).

Applications:

- **Chatbots** → ChatGPT, Claude.
- **Code Generation** → Copilot, Code Llama.
- **Summarization/Translation** → Summarize articles, translate languages.

Relations in One Flow

1. **AI** → Big umbrella, the dream of smart machines.
2. **Data Science** → Uses AI/ML + statistics for practical insights.
3. **ML** → A core method of AI → teaches machines from data.
4. **DL** → Advanced ML using neural networks.
5. **ANN/CNN/RNN/LSTM** → Specialized architectures for different tasks (vision, sequences).
6. **Transformers** → Revolution in sequence learning → foundation of modern NLP.
7. **LLMs** → Gigantic Transformer models trained on huge datasets → today's Generative AI.

Datasets for Machine Learning

Please refer below links of datasets that we can use to solve case studies.

In Case study round almost below datasets are referred.

Open Datasets

Kaggle

A data science community with tools and resources which include externally contributed machine learning datasets of all kinds. From health, through sports, food, travel, education, and more, Kaggle is one of the best places to look for quality training data.

Google Dataset Search

A search engine from Google that helps researchers locate freely available online data. It works similarly to Google Scholar, and it contains over 25 million datasets. You can find here economic and financial data, as well as datasets uploaded by organizations like WHO, Statista, or Harvard.

OpenML

An online machine learning platform for sharing and organizing data with more than 21.000 datasets. It's regularly updated and it automatically versions and analyses each dataset and annotates it with rich meta-data to streamline analysis.

DataHub

A collection of thousands of machine learning datasets from financial market data, macroeconomic data, and population growth to cryptocurrency prices. You can access it without any registration.

Papers with Code

A community project with free and open resources, currently including 3937 datasets for data science and machine learning, including natural language processing tasks. You can easily filter them by modality, task, or language.

VisualData

A search engine for computer vision datasets. You can easily filter them by category, date, popularity or use a search box to find a theme-specific dataset. A great source of datasets for image classification, image processing, and image segmentation projects.

Public Government datasets

Data.gov

The US government's open data site. You can filter it by various industries like healthcare, climate, education, etc. Be aware that much of this open-source data might require additional research.

EU Open Data Portal

The point of access to public data published by the EU institutions, agencies, and other entities. It contains data related to economics, agriculture, education, employment, climate, finance, science, etc.

World Bank

The open data from the World Bank that you can access without registration. It contains data concerning population demographics, macroeconomic data, and key indicators for development. A great source of data to perform data analysis at a large scale.

US Healthcare Data

Statistics and datasets for health care and public health. You can find data about population health, diseases, drugs, and health plans collected from the FDA and USDA Food composition databases.

The US National Center for Education Statistics

It's a website with data on educational institutions and education demographics in the U.S. and internationally.

The UK Data Service

It's a platform that provides access to over 7,000 digital data collections for research and teaching purposes. You can find here economic and social data from the Economic and Social Data Service (ESDS), Census Programme, and others, including some international data sets.

Data USA

A free platform with the most comprehensive visualization of U.S. public data.

Machine Learning Datasets for Finance and Economics

Global Financial Development (GFD)

An extensive dataset of financial system characteristics for 214 economies around the world. It contains annual data which has been collected since 1960.

Financial Times Markets Data

Up-to-date source of data on financial markets from around the world. The dataset contains information about share and stock prices, equities, currencies, bonds, and commodities performance.

Quandl

A platform with rich datasets of financial, economic, and alternative data. Quandl's data comes in two formats: Time-series (data taken over a period of time) and Tables (numerical and unsorted data types such as strings, etc.) You can download it either as a JSON or CSV file.

IMF Data

International Monetary Fund publishes data related to the IMF lending, exchange rates, and other economic and financial indicators.

American Economic Association (AEA)

A website with links to some of the most useful and popular economic data sources. It includes data on U.S macroeconomic as well as individual-level global data on income, employment, and health.